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Study on spatio-temporal patterns of gray matter and white matter functional connectivity in autism spectrum disorder based on Mamba-CNN framework

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Abstract

Purpose Autism Spectrum Disorder (ASD) is a neurodevelopmental disorder that widely affects children and adults. Early and accurate diagnosis is crucial for improving patient outcomes, but traditional approaches face challenges in capturing the complex spatio-temporal dynamics of brain networks.

Methods This study included multi-center data from ABIDE I and ABIDE II, comprising resting-state functional magnetic resonance imaging and structural magnetic resonance imaging data of 1175 participants. To address the technical challenges in analyzing neuroimaging data, we proposed an innovative deep learning framework based on the Mamba-CNN architecture, which leverages selective state space mechanisms to capture dynamic interaction features and long-range dependencies. Additionally, a Hierarchical Multi-scale Fusion module was introduced to integrate brain network features across different spatiotemporal scales.

Results We systematically investigated the specific features of dynamic network connectivity between gray matter and white matter in ASD populations for the first time, identifying unique abnormal patterns such as enhanced connectivity between the superior longitudinal fasciculus and dorsal anterior cingulate cortex, and reduced connectivity between the corpus callosum and somatosensory association cortex. Experimental results demonstrated that this framework achieved an 87.49% accuracy, 87.31% sensitivity, 87.68% specificity in ASD diagnosis, outperforming other ASD classification methods.

Conclusions This study not only enriches the understanding of ASD's neural mechanisms from the perspective of gray matter and white matter dynamic connectivity but also provides a new technical solution for neuroimaging data analysis.

Keywords Autism spectrum disorder · Gray matter · White matter · Spatio-temporal patterns · Mamba-CNN

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Introduction

Autism Spectrum Disorder (ASD) is a complex neurodevelopmental disorder characterized by core clinical features such as impairments in social communication, restricted and repetitive behaviors, and narrow interests [1]. With societal development and increased awareness of mental health, ASD has gradually emerged as a significant public health concern [2]. Global epidemiological data indicate a consistent upward trend in the prevalence of ASD. Although regional variations exist in specific incidence rates, the overall growth pattern is notably prominent [3]. This condition not only severely impacts patients' quality of life and independence but also substantially increases the economic and caregiving burdens on families and society, as manifested by elevated healthcare expenditures and expanded demands for special education [4]. In this context, achieving early and accurate diagnosis of ASD is of paramount importance, early diagnosis enables timely access to individualized interventions, thereby effectively ameliorating clinical symptoms and long-term prognosis, improving the quality of life for both patients and their families, and simultaneously alleviating comprehensive societal burdens [5].

To accurately identify biomarkers of ASD and advance its diagnostic techniques, researchers have conducted systematic investigations based on structural magnetic resonance imaging (sMRI) and functional magnetic resonance imaging (fMRI) data. Current studies primarily focus on the construction and analysis of gray matter cortical functional networks, aiming to elucidate the neurobiological mechanisms underlying ASD [6]. Existing evidence demonstrates significant differences between individuals with ASD and healthy controls (HCs) in terms of gray matter cortical volume, density, and functional connectivity patterns [7]. For instance, Ma et al. analyzed sMRI data and confirmed that individuals with ASD exhibit reduced gray matter volume in the bilateral cerebellar hemispheres, right parahippocampal gyrus, right orbitofrontal cortex, and right anteroventral thalamic nucleus [8]. Meanwhile, several studies have also emphasized the importance of structural similarity networks derived from multiple morphological features. For instance, Zheng et al. extracted seven morphological features from 360 cortical regions and employed an elastic network to quantify cortico-cortical similarities, constructing multi-feature-based networks for ASD classification [9]. In addition to gray matter, white matter serves as a critical neural fiber pathway connecting distinct brain regions, playing a pivotal role in efficient information transmission and brain functional integration [10, 11]. In recent years, researchers have increasingly focused on the potential contributions of white matter characteristics to the pathophysiological mechanisms of ASD. Some studies have revealed impaired white matter integrity in children with ASD, particularly in white matter fiber tracts associated with the frontoparietal-temporal-occipital functional circuits [12]. Other evidence suggests that alterations in the structure and function coupling of white matter regions are correlated with the severity of clinical symptoms in ASD [13]. However, the construction of brain functional networks inherently relies on the synergistic interaction between gray matter and white matter [14]. Although existing research has identified abnormalities in either gray matter or white matter as isolated modalities, the gray matter and white matter functional network connectivity, as the core pathway for interregional information transfer, may play a more critical role in the pathogenesis of ASD. Systematic investigations into gray matter and white matter functional network connectivity not only address existing gaps in the understanding of ASD neurobiology but also hold promise for providing innovative perspectives toward the development of highly sensitive diagnostic biomarkers, thereby offering substantial value for both scientific research and clinical translation.

Currently, the analysis of functional network data involving gray matter and white matter poses multiple challenges. Firstly, the dynamic interaction patterns are highly complex, the global functional activities between gray matter and white matter not only exhibit spatiotemporal dynamic characteristics, but also the brain functional network patterns among different subjects show high heterogeneity. Secondly, traditional analytical methods struggle to capture the laws of their transient fluctuations and long-range dependencies [15, 16]. The rapid advancement of Artificial Intelligence (AI) technology has provided a novel paradigm for addressing the aforementioned challenges [17]. As a core subfield of AI, deep learning through multi-layer nonlinear transformations and end-to-end training can automatically extract multi-level features from massive, high-dimensional, and heterogeneous

neuroimaging data, thereby overcoming the limitation of traditional machine learning, which relies heavily on manual feature engineering [18].

The Mamba algorithm, as an emerging deep learning architecture, offers the advantage of efficiently modeling and extracting features from high-dimensional sequence data through State Space Models (SSMs) [19–21]. Compared to traditional Recurrent Neural Networks, the Mamba algorithm dynamically regulates information flow via a Selective State Space mechanism, enabling more precise capture of dynamic interaction features and long-range dependencies [22]. This characteristic endows it with significant potential in processing biomedical data characterized by complex spatio-temporal dynamics [23, 24]. Based on the aforementioned research, this study proposes a Mamba-CNN deep learning framework, which explores the distinctive characteristics of dynamic network connectivity between gray matter and white matter in ASD, thereby providing technical support for the early and precise diagnosis of ASD. Overall, the main contributions of our work can be summarized as follows:

[1]. This study is the first to systematically investigate the specific characteristics of dynamic network connectivity between gray matter and white matter in ASD, offering a new structural-functional connectivity perspective for elucidating the neural mechanisms of the disorder.
[2]. This study proposes a deep learning framework based on the Mamba-CNN architecture, which accurately captures the dynamic interaction characteristics of brain networks and long-range dependencies through the selective mechanism of state space models.
[3]. We introduce a Hierarchical Multi-scale Fusion module to effectively integrate brain network features across different spatio-temporal scales, enhancing the model's ability to represent complex neural connectivity patterns.

Materials and methods

Participants

Resting-state fMRI and sMRI data used in this study were obtained from the open-access databases ABIDE I [25] and ABIDE II [26]. A total of 15 sites' data were included in this study, with the inclusion criterion that the repetition time (TR) for functional imaging data acquisition at all these sites was uniformly 2000 ms. Excluded subjects whose imaging data shows some missing. Finally, data of 607 ASD individuals and 568 HCs was included in our study. Table 1 shows the cohort details, including distribution of ASD and HCs by sex, age, and the mean frame-wise displacement (FD) quality measure [27]. There was no difference in the age, sex and FD between cases and healthy controls ($p > 0.05$).

Table 1 Demographics and clinical characteristics of the participants

SITE ID	ASD			HCs		
	Age	Gender	FD	Age	Gender	FD
CALTECH	24.75±7.82	M 11, F 3	0.07	26.7±8.86	M 10, F 4	0.07
CMU	24.67±4.87	M 8, F 1	0.34	26.73±6.18	M 9, F 2	0.32
NYU	13.92±6.13	M 159, F 22	0.12	14.30±5.97	M 106, F 27	0.08
SDSU	14.62±5.92	M 36, F 8	0.11	14.57±5.89	M 36, F 8	0.07
STANFORD	13.55±6.05	M 35, F 6	0.11	13.57±6.03	M 35, F 6	0.11
TRINITY	17.28±3.57	M 24, F 0	0.13	17.23±3.90	M 23, F 0	0.10
UM	13.88±3.29	M 58, F 10	0.19	14.81±3.35	M 59, F 18	0.09
USM	22.65±7.73	M 58, F 0	0.16	21.36±7.64	M 43, F 0	0.10
YALE	12.69±3.09	M 19, F 8	0.12	12.68±2.75	M 20, F 8	0.09
EMC	8.04±1.19	M 22, F 5	0.13	8.15±0.99	M 22, F 5	0.10
ETH	20.56±3.40	M 13, F 0	0.13	23.88±4.50	M 24, F 0	0.08
GU	10.85±1.52	M 42, F 8	0.14	10.47±1.70	M 28, F 26	0.09

Table 1 Demographics and clinical characteristics of the participants

SITE ID	ASD			HCs		
	Age	Gender	FD	Age	Gender	FD
TCD	15.03±3.15	M 20, F 0	0.15	15.71±3.16	M 20, F 0	0.09
UCD	14.75±1.97	M 14, F 4	0.14	14.80±1.71	M 10, F 4	0.11
UMIA	9.98±2.01	M 11, F 2	0.14	9.79±2.09	M 11, F 4	0.12
ALL Sites	14.24±6.44	M 530, F 77	0.16	14.10±6.32	M 456, F 112	0.10

Abbreviations: ASD, autism spectrum disorder; HCs, healthy controls; M, male; F, female, FD, frame-wise displacement

MRI data preprocessing

The rs-fMRI data were preprocessed using the statistical parametric mapping software package SPM12 (http://www.fil.ion.ucl.ac.uk/spm). The following steps were performed: (1) removal of the first 10 time-points; (2) slice-timing correction; (3) head-movement correction; (4) registration; (5) skull stripping; (6) spatial normalization; (7) spatial smoothing; (8) detrending; (9) regression covariates; (10) filtering; (11) segmentation. We segmented the brain into gray matter and white matter based on T1 weighted images and regressed the three according to the actual situation. The gray matter mask included 82 regions directly from the Brodmann atlas and the white matter bundle mask consisted of 48 regions based on the JHU-MNI-ss atlas [28]. Extracted region of interest (ROI) time series from both the gray matter and white matter masks were used for subsequent analysis.

Methods

After performing the standard preprocessing of rs-fMRI data and segmenting ROIs in gray matter and white matter, this study employed the sliding window method to investigate the dynamic functional connectivity and spontaneous activity within and between gray matter and white matter regions. As shown in Fig. 1(a), the gray matter and white matter dynamic functional network connectivity matrices were constructed by calculating Pearson correlation coefficients of time series within each window, capturing temporal variations in brain network dynamics. Then we proposed the Mamba-CNN model for analyzing the spatio-temporal patterns of gray matter and white matter functional connectivity in ASD (Fig. 1(b)). In our Mamba-CNN model, the encoder adopts a Mamba architecture, utilizing a four-directional scanning strategy for sequence modeling to capture the spatial information of the input gray matter and white matter dynamic functional network connectivity matrices. The decoder, along with skip connections, follows a CNN structure, employing progressive convolution to extract and restore feature information, thereby refining the features of the gray matter and white matter dynamic functional network connectivity matrices. The specific network architecture is illustrated in Fig. 1. During the analysis, time-varying gray matter and white matter dynamic functional network connectivity matrices are fed into the network as input. The encoder extracts dynamic feature information from these matrices, which is then passed to the decoder for upsampling to generate the predicted output. Throughout the feature propagation process, a multi-scale spatial-channel fusion strategy is employed to extract and integrate channel-wise and spatial attributes across different scales. Finally, the decoder outputs the probability of the subject being diagnosed with ASD.

Encoder

The gray matter and white matter functional connectivity exhibits significant temporal dynamic characteristics. To effectively extract and model the latent state transition patterns between adjacent temporal frames, the encoder adopts a SSMs architecture that employs long-range temporal dependency modeling to characterize the state transition dynamics in brain functional connectivity time series. The detailed network architecture is illustrated in Fig. 2. The encoder architecture consists of three components: the Patch Embedding module, the Visual State Space (VSS) block, and the Patch Merging module. Among these, the Patch Embedding and Patch Merging modules perform downsampling operations to reduce feature spatial resolution and computational complexity.

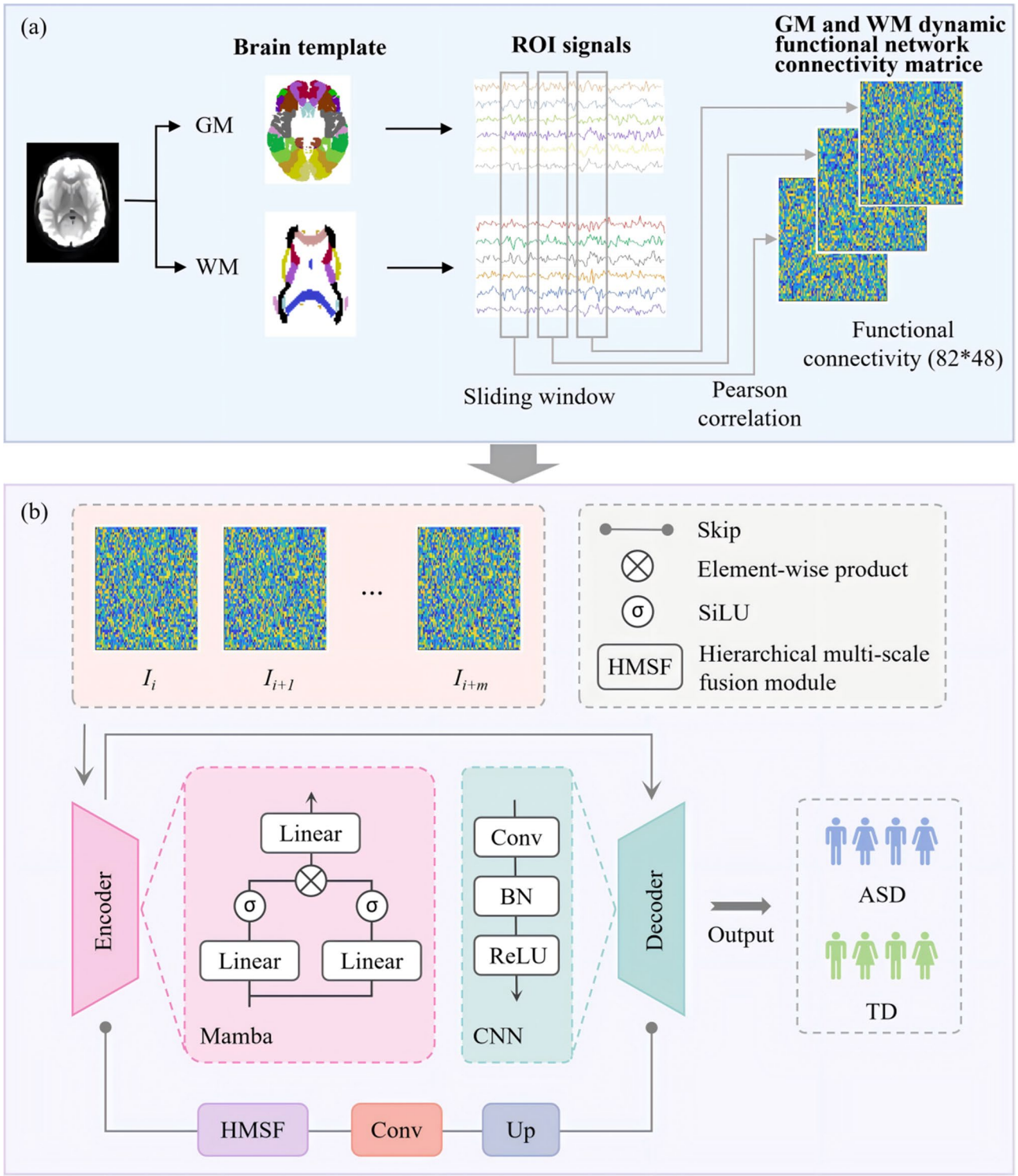


Fig. 1 The pipeline of the current study

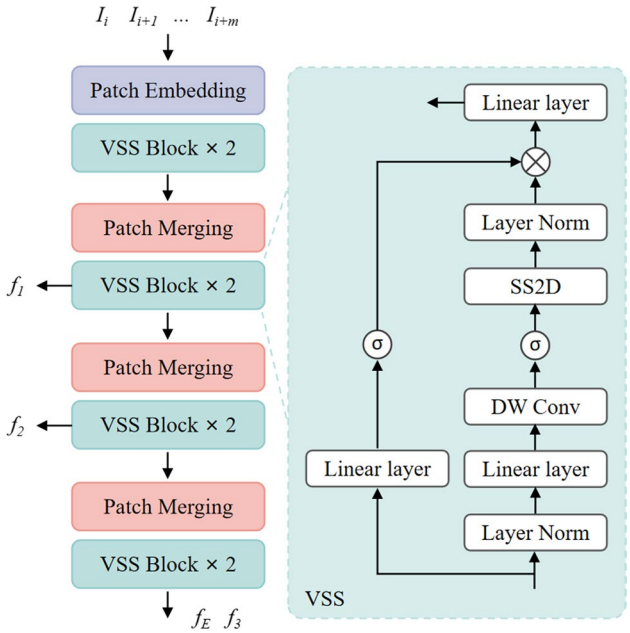


Fig. 2 Network structure of Encoder

Specifically, the encoder input is formed by concatenating m consecutive frame matrices along the channel dimension, generating a composite feature representation with dimensions of $m\times3$. During feature extraction, the input features undergo four progressive downsampling stages ($\frac{1}{4},\frac{1}{8},\frac{1}{16},\frac{1}{32}$), ultimately producing multi-scale feature representations f_E. Notably, the features f_1, f_2, and f_3 generated from the 2nd, 3rd, and 4th downsampling stages are selectively retained and transferred to subsequent network modules as inputs for skip connections.

The VSS block represents a vision-specific adaptation of the Mamba model, serving as the core component for global feature extraction within the encoder. The architectural details of the VSS block are illustrated in Fig. 2. The VSS block employs 2D Selective Scan (SS2D) to process visual data: (1) it unfolds the input image into four distinct spatial sequences by scanning along four directional paths (left to right, top to bottom, right to left, and bottom to top), thereby encoding diverse spatial contextual information; (2) it extracts and fuses features from these sequences to reconstruct the original spatial resolution. The feature extraction process in SS2D operates independently across the four generated sequences. Each sequence undergoes selective state space processing via the S4 mechanism to filter irrelevant information and retain task-relevant features. The computational formulation is expressed as follows:

$$h_t=\overline{A_{t-1}}h_{t-1}+\overline{B}_t x_t \tag{1}$$

$$y_t=G_t h_t \tag{2}$$

In the formula, $A\in R^{N\times N}$ denote the state transition matrix, $B,G\in R^N$ represent the projection parameters, and N indicate the state dimensionality. h corresponds to the hidden state that maps the input excitation x to the output response y. Among them, $\overline{A}_t=\exp(A_t\Delta_t)$, $\overline{B}_t=dB_t\Delta_t$, $\Delta_t=dx^t_{t-1}$. At this stage, four feature sequences with distinct spatial properties are obtained. These sequences undergo scanning-based merging and reshaping to reconstruct the original image representation for subsequent propagation through the network. Subsequently, Layer Normalization is applied to regularize the feature distribution output by SS2D. Finally, a linear layer with residual connection is employed to fuse the multiscale feature information.

Hierarchical multi-scale fusion

The gray matter and white matter functional connectivity network exhibits rapid temporal dynamics while undergoing topological reconfiguration across brain regions in the spatial dimension. This coupled spatio-temporal variation necessitates that models learning the gray matter and white matter functional connectivity patterns must not only capture fast dynamic interactions but also comprehend the evolving topological restructuring of the brain network. The multi-scale nature of the gray matter and white matter functional connectivity network poses challenges for traditional single-scale modeling approaches, which struggle to effectively distinguish subtle differences between ASD patients and healthy controls. To focus the model's attention on extracting the intrinsic characteristics of the gray matter and white matter functional connectivity network, a Hierarchical Multi-scale Fusion (HMSF) strategy is proposed. This strategy integrates feature information across different temporal and spatial scales, mitigating the interference of scale variations in the gray matter and white matter functional connectivity network on the model's diagnostic performance, thereby achieving more precise identification of ASD. The schematic of the proposed strategy mechanism is illustrated in Fig. 3.

The HMSF module comprises two primary components: the channel group and the spatial group, which integrate feature information across three distinct scales (1×1, 3×3, and 5×5). The specific fusion process is as follows: the input matrix of temporal frames, after undergoing a series of encoder convolutional operations, generates feature representation $f_a \in R^{H \times W \times C}$, H,W,C denote the height, width, and number of channels of the feature map, respectively. Here, a=1,2,3. This feature f_a is then propagated through both the channel group and the spatial group within the HMSF module. Consequently, it yields feature $f_C \in R^{H \times W \times C}$ with channel-wise weighting and feature $f_S \in R^{H \times W \times C}$ with spatial-wise weighting. Within the spatial group, feature f_a undergoes two consecutive 3×3 convolutional operations. Specifically, during the first convolution, the feature map $f_{3\times3}$ exhibits a receptive field of 3×3, following the second convolution, the resulting feature map $f_{5\times5}$ demonstrates an expanded receptive field of 5×5. Subsequently, pooling operations are applied to the two feature maps with distinct receptive fields along the spatial dimension, generating two spatial feature representations with channel dimension $1 \times 1 \times C$. These two spatial features undergo channel-wise scaling and summation to integrate multi-receptive-field spatial information, thereby facilitating cross-channel spatial interaction. Finally, the Sigmoid function is applied to transform the integrated spatial features into spatial weight maps M_S, which are then element-wise multiplied with f_a to produce the refined spatial feature f_S. Within the pooling group, f_a is processed through two

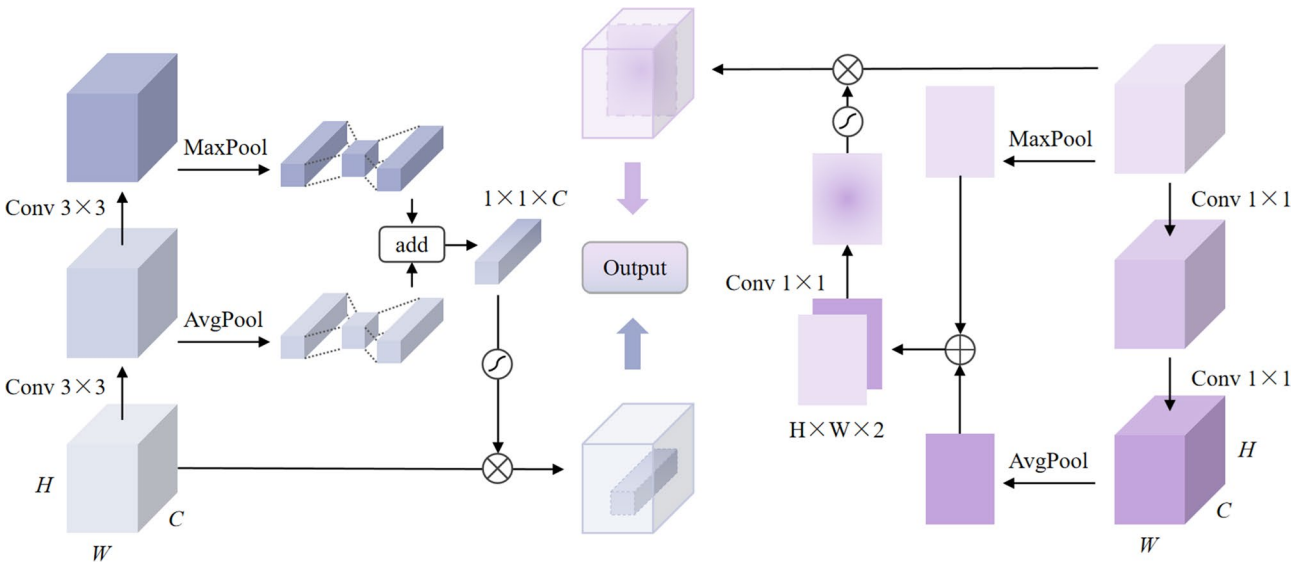


Fig. 3 Hierarchical Multi-scale Fusion mechanism diagram

consecutive 1×1 convolutional operations to generate the feature $f_{1\times1}$ with a 1×1 receptive field. Subsequently, pooling operations are applied to features f_a and $f_{1\times1}$ along the channel dimension, generating two channel-wise feature representations with spatial dimension $H \times W \times 1$. These two channel features are then concatenated and processed through a convolutional operation to produce the channel-fused feature representation. Finally, the channel feature information is transformed into channel weights M_C via the Sigmoid function. The weights M_C are then applied to weight the feature f_a, yielding the refined feature f_C. The final output of the HMSF module is obtained by concatenating the spatially weighted feature f_S and the channel-weighted feature f_C along the channel dimension. Consequently, the computational process of HMSF is formally expressed by Eq. (5).

$$M_S = \sigma\{\mathrm{MLP}(\mathrm{AvgPool}(f_{3\times3})) + \mathrm{MLP}(\mathrm{MaxPool}(f_{5\times5}))\} \tag{3}$$

$$M_C = \sigma\left\{\mathrm{conv}\left[\mathrm{cat}\begin{pmatrix}\mathrm{Avgpool}(f_{1\times1}),\\ \mathrm{Maxpool}(f_a)\end{pmatrix}\right]\right\} \tag{4}$$

$$F_{HMSF}(f_a) = \mathrm{cat}((f_a \times M_C),(f_a \times M_S)) \tag{5}$$

Here, $\sigma\{\cdot\}$ denotes the channel weights obtained by applying the Sigmoid activation function to transform feature information into values within the range of [0, 1], $cat(\cdot,\cdot)$ represents the channel-wise concatenation operation; conv indicates the 1×1 convolution operation, multi-layer perceptron (MLP) refers to a two-layer convolutional structure that achieves channel scaling of the input features, Avgpool denotes the average pooling operation and Maxpool represents the maximum pooling operation.

Decoder

In the encoder, the SS2D module exhibits the capability to filter irrelevant information. To compensate for the potential loss of fine-grained features caused by SS2D, the decoder employs classical convolutional operations to restore the feature representations extracted by the encoder. Simultaneously, skip connections are implemented between hierarchical layers of the encoder's state space to preserve local feature information in a timely manner. The decoder and encoder adopt an asymmetric architectural design. Since the encoder reduces the spatial resolution of the input image by a factor of 32, the decoder incorporates a progressive feature reconstruction process comprising five upsampling operations ($\frac{1}{16},\frac{1}{8},\frac{1}{4},\frac{1}{2},1$). The detailed network architecture is illustrated in Fig. 4.

The decoder input comprises two components: the encoder output f_E and the skip connection k_a. The decoder outputs the frame-level probability of ASD diagnosis for each temporal frame via progressive upsampling and skip connection fusion. The skip connection is derived from the encoder output f_1,f_2,f_3 through a series of CNN operations. The transformation process is mathematically formulated as follows:

$$\mathrm{out}_a = \mathcal{F}_{HMSF}(f_a) \tag{6}$$

$$k_a = \mathrm{Up}\{\mathrm{ReLU}[\mathrm{BN}(\mathrm{Conv}(\mathrm{out}_a))]\} \tag{7}$$

Among them, a=1,2,3. $\mathrm{up}(\cdot)$ denotes the upsampling operation that doubles the spatial dimensions (height and width) of the feature map. $\mathrm{ReLU}[\cdot]$ represents a nonlinear activation function. $\mathrm{BN}(\cdot)$ indicates the regularization process, and $\mathrm{conv}(\cdot)$ refers to the 3×3 convolutional operation. $F_{HMSF}(f_a)$ is the HMSF operation. f_a represents the shallow features extracted by the encoder. Shallow features typically encapsulate abundant micro-scale information, including local gray matter and white matter connectivity strength, short-range functional interaction patterns, and time-varying signal fluctuations. These high-frequency dynamic features are prone to attenuation or loss during the multi-layer convolutional processes in deep networks due to spatial smoothing operations. By implementing adaptive weight preservation, cross-temporal feature alignment, and hierarchical skip connections for these shallow features, and precisely integrating them into corresponding resolution levels of the decoder, the

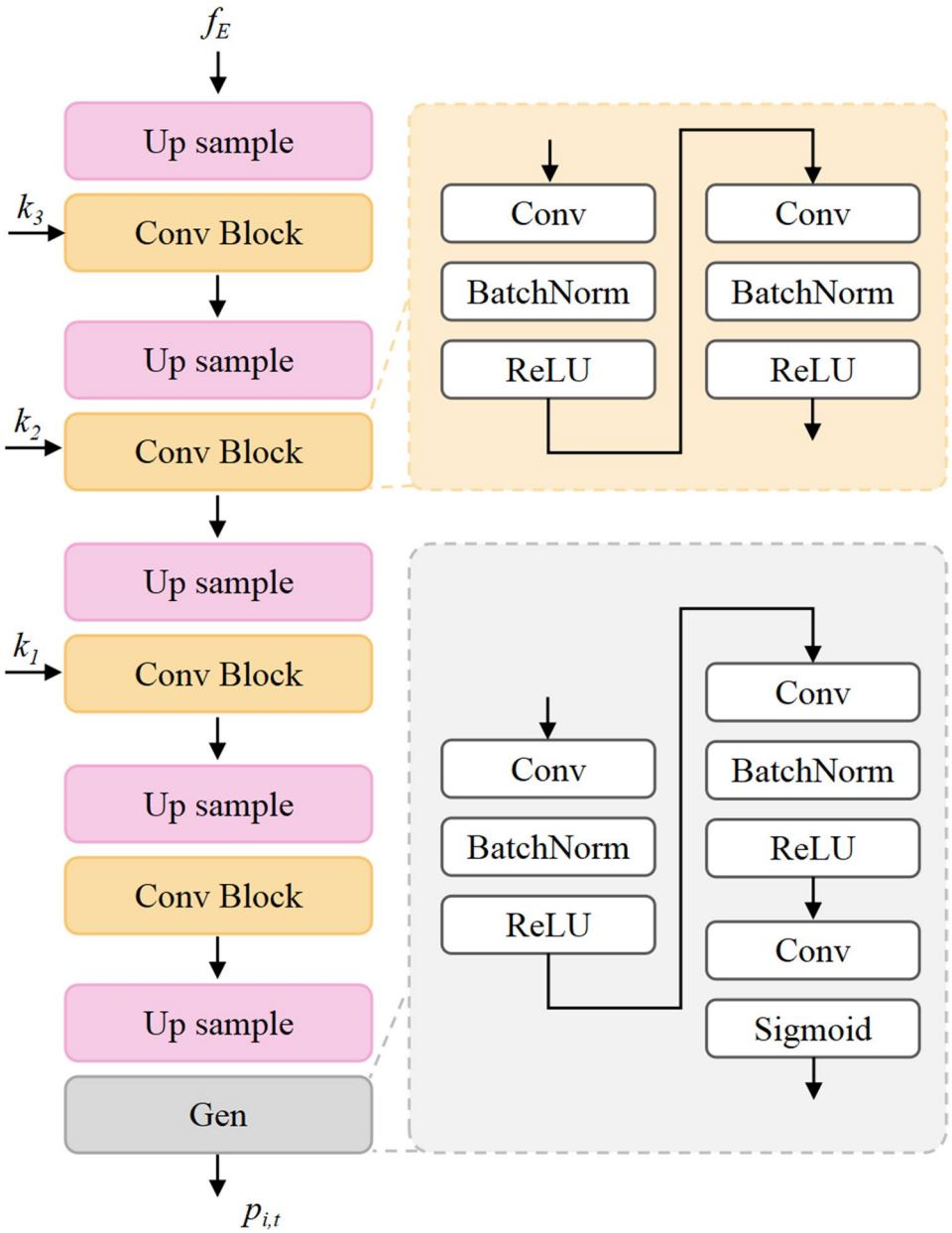


Fig. 4 Network structure of the Decoder

approach enables effective fusion of whole-brain network topological architecture with local functional connectivity patterns. This strategy constrains the dynamic functional connectivity reconstruction process, progressively restoring the complete feature representation of functional connectivity matrices from low-rank approximations to original high-dimensional spatio-temporal resolution, thereby enhancing the sensitivity of ASD diagnosis and the spatial precision of functional abnormality localization. Finally, a majority voting mechanism is employed to consolidate the sequence of frame-level ASD probability estimates produced by the decoder into a single subject-level diagnosis, ensuring robustness against transient noise by selecting the modal class across the temporal dimension.

Loss function

In this study, we employ the Binary Cross Entropy (BCE) loss to supervise the frame-level ASD classification task. The loss directly optimizes the model to discriminate between ASD and HCs based on dynamic gray matter and white matter functional connectivity patterns within each sliding-window instance. The computational formulation is presented as follows:

$$\mathcal{L}_{BCE}=-\frac{1}{N \cdot T}\sum_{i=1}^{N}\sum_{t=1}^{T}[y_i\log(p_{i,t})+(1-y_i)\log(1-p_{i,t})] \tag{8}$$

Among them, N is the number of samples, T is the number of frames, y_i denotes the real label of the *i*-th sample, $p_{i,t}$ denotes the model's predicted probability that the gray matter and white matter dynamic functional connectivity matrix of the *i*-th sample at the *t*-th frame belongs to the positive class. By minimizing this loss, the model learns to assign higher probabilities to window instances from ASD and lower probabilities to those from HCs. This frame-level supervision leverages the rich temporal dynamics of rs-fMRI data while remaining strictly aligned with the ultimate subject-level classification objective.

Experiments and results

Experiments setup

In this section, the diagnostic efficacy of the proposed ASD classification framework was systematically assessed. Model training and evaluation were conducted using PyTorch framework with GPU acceleration to enhance computational efficiency. The implementation environment consisted of an NVIDIA Quadro M4000 GPU paired with an Intel Core i7 processor operating at 2.4 GHz, ensuring optimized performance for deep learning workloads. In this study, the learning rate for Mamba-CNN model was set at 0.001, with a dropout rate of 0.3 and a maximum of 600 training iterations.

Benchmark

To confirm the efficacy of the proposed Mamba-CNN method, we compared it with other competing approaches, including VGG [29], ResNet [30], DenseNet [31], and Transformer [32]. All models were trained with the same input of brain gray matter and white matter dynamic functional connectivity information. The number of the hyperparameters for each method were chosen to match those specified in the respective original papers.

Evaluation metrics

The metrics used to evaluate model performance were accuracy (ACC), sensitivity (SEN), specificity (SPE), area under the curve (AUC) of receiver operating characteristic (ROC) and F1 score. Binary classification metrics were evaluated based on the optimal cut-off value determined by the maximum Youden index on the ROC curve [33–35]. The DeLong test is employed for statistical comparison between the proposed model and other approaches.

Classification results based on different window lengths

This study employs dynamic functional connectivity to capture the time-varying coupling characteristics of gray matter and white matter, where the appropriate selection of sliding window length is a critical parameter to ensure the validity of the analysis. Specifically, while longer windows can yield more stable functional connectivity

results, they tend to obscure subtle fluctuations in dynamic network changes and diminish the ability to detect temporal dynamic features. Conversely, shorter windows enhance sensitivity to dynamic changes but reduce statistical power and increase the risk of noise introduction, leading to decreased stability in functional connectivity [36]. In the existing studies, Zhuang et al. [37] determined the optimal sliding window length to be 20 time points (corresponding to TR = 2000 ms) using the grid search method. In contrast, Cui et al. [38] identified the best window length as 5 time points. However, due to differences in TR across data acquisition sites in different studies, the actual temporal duration corresponding to the window length varies. Ren et al. [39] set the sliding window length to 30 time points (TR = 2000 ms / 3000 ms). Based on the above research foundation and combined with the sampling rate of TR = 2000 ms in our study's data, the window time length was controlled within the range of 30–60 s, corresponding to setting 15–30 time points as one window. Therefore, we first explored the classification results when the sliding window lengths were 15, 20, 25, and 30 time points respectively, with the sliding window stride is 1. The window length was determined using a separate 5-fold cross-validation on a randomly selected 80% of the entire cohort, with the remaining 20% used only for validation. Candidate window lengths of 15, 20, 25, and 30 time points were compared, and the length yielding the highest average validation AUC was selected and fixed. As displayed in Fig. 5, the experimental results showed that when the sliding window length was 20 time points, the model achieved the optimal performance in distinguishing ASD from HCs, with the classification accuracy and related indicators reaching the highest level. Crucially, to prevent data leakage due to the high overlap of sliding windows, all cross-validation splits were performed strictly at the subject level before dynamic feature extraction. The dataset was partitioned into training and test subsets with no subject overlap. Subsequently, the sliding window algorithm was applied independently to each subset to generate dynamic functional network connectivity matrices. This ensured that all dynamic functional network connectivity samples within the training set originated from distinct participants than those in the test sets.

Results comparison

To assess the generalizability of the Mamba-CNN model to unseen data collected from different study sites, a leave-one-site-out analysis was conducted by iteratively using the data from 14 study sites to train the model, and the data from the left-out site for testing the model (Table 2). The model performance of performing leave-one-site-out is comparable to that of using cross-validation across all sites. For each leave-one-site-out fold, the classification threshold was determined using 5-fold cross-validation exclusively within the 14 training sites, without access to the held-out test site. Specifically, we evaluated thresholds ranging from 0 to 1 at 0.01 intervals and selected the threshold that maximized the average Youden index across the 5 validation folds. This threshold was

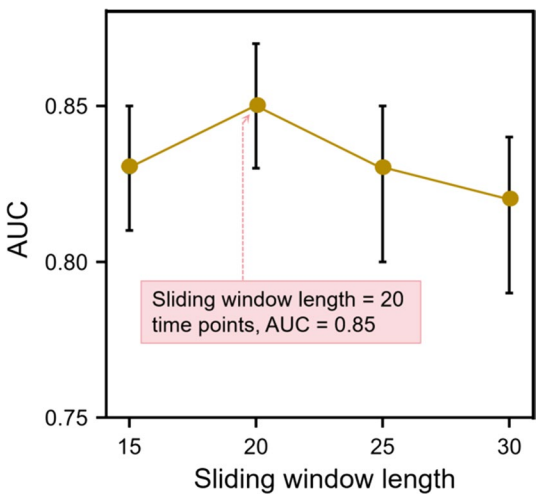


Fig. 5 The overall classification AUC for different window lengths ranging from 15 to 30 time points

Table 2 Classification performance with various measurements

SITE ID	AUC (95% Confidence interval)	ACC (%)	SEN (%)	SPE (%)	F1 score	*P* value
CALTECH	0.81 (0.61–0.93)	85.71	85.71	85.71	0.86	<0.05
CMU	0.86 (0.63–0.97)	85.00	88.89	81.82	0.84	<0.05
NYU	0.90 (0.86–0.93)	90.13	88.95	91.73	0.91	<0.05
SDSU	0.88 (0.79–0.94)	87.50	86.36	88.64	0.87	<0.05
STANFORD	0.87 (0.78–0.93)	86.59	87.80	85.37	0.87	<0.05
TRINITY	0.90 (0.78–0.97)	87.23	87.50	86.96	0.88	<0.05
UM	0.85 (0.78–0.90)	84.14	82.35	85.71	0.83	<0.05
USM	0.88 (0.80–0.94)	91.09	91.38	90.70	0.92	<0.05
YALE	0.81 (0.68–0.90)	85.45	85.19	85.71	0.85	<0.05
EMC	0.86 (0.74–0.94)	87.04	88.89	85.19	0.87	<0.05
ETH	0.87 (0.72–0.96)	86.49	84.62	87.50	0.81	<0.05
GU	0.84 (0.76–0.91)	85.58	86.00	85.19	0.85	<0.05
TCD	0.83 (0.68–0.93)	85.00	85.00	85.00	0.85	<0.05
UCD	0.83 (0.66–0.94)	87.50	88.89	85.71	0.89	<0.05
UMIA	0.83 (0.64–0.94)	85.71	84.62	86.67	0.85	<0.05
All sites	0.85 (0.83–0.87)	87.49	87.31	87.68	0.88	<0.05

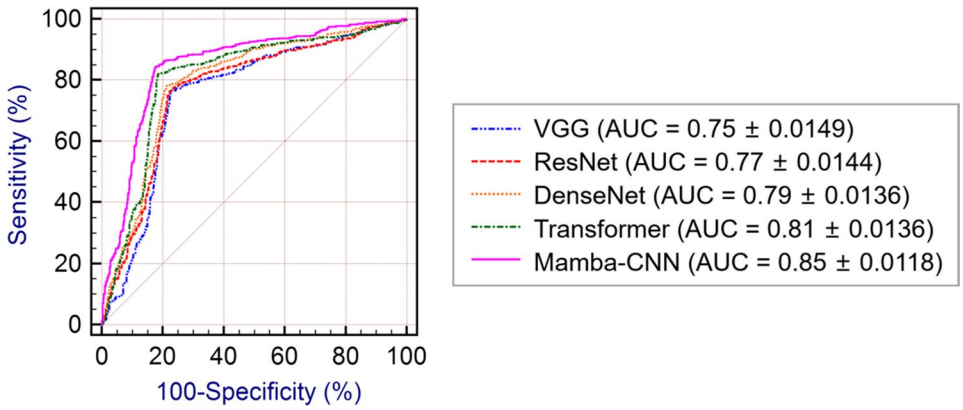


Fig. 6 The ROC curves comparing our method with benchmark methods

then fixed for evaluating the held-out test site in that fold. The results demonstrated the robustness of our model for unseen data from different study sites.

Figure 6 summarizes the ROC results of ASD diagnosis for all the compared methods. The overall results show that Mamba - based model surpass the traditional deep learning methods. Importantly, our proposed Mamba-CNN yields the best results for all the performance metrics (87.49% accuracy, 87.31% sensitivity, and 87.68% specificity). There was a statistically significant difference in classification performance between Mamba-CNN and VGG, ResNet, DenseNet respectively, with the significance levels (P-values) of the DeLong test being 0.0001, 0.0114, and 0.0102. In contrast, no statistically significant difference in classification performance was observed between Transformer and Mamba-CNN, and the significance level (P-value) of the DeLong test was 0.2805.

Additionally, we compared our proposed framework, against several existing state-of-the-art models. The performance metrics for each model are summarized in Table 3. These findings demonstrate that Mamba-CNN outperforms existing advanced models such as kSVM [40], Site-Adaptive [41], MADE-for-ASD [42], ASD-GraphNet [43], GCN [44], and BrainTGL [45]. Although the MMEC method [46] reached 97.82% accuracy through ensemble deep learning, several key differences should be considered. Firstly, unlike our study's comprehensive ABIDE I and II dataset, MMEC relied on partial subsets from both ABIDE I and II, which may artificially inflate performance metrics due to reduced data diversity. Moreover, while MMEC implements a CNN ensemble approach, its lack of explainable artificial intelligence components restricts clinical interpretability, which is a critical requirement for medical applications.

Table 3 Comparison with state-of-the-art methods for ASD classification

Method	Sample size	AUC	ACC (%)	SEN (%)	SPE (%)	F1 score
SVM [15]	107	-	72.94	74.70	70.83	-
kSVM [40]	871	-	69.43	64.57	73.61	-
Site-Adaptive [41]	1035	-	73.00	-	-	-
MADE-for-ASD [42]	1035	-	75.20	82.90	69.70	-
ASD-GraphNet [43]	1035	-	75.25	75.07	76.40	0.75
EV-GCN [47]	870	0.77	71.30	-	-	-
S3TL [44]	1035	0.73	69.14	67.27	70.88	-
BrainTGL [45]	1112	0.68	67.80	62.20	72.90	-
MMEC [46]	569	-	97.82	97.43	98.22	0.98
Our method	1175	0.85	87.49	87.31	87.68	0.88

Table 4 The ablation results across these different connectivity representations and model components

Grouping	Input features	HMSF component	AUC (95% Confidence interval)	ACC (%)	SEN (%)	SPE (%)	F1 score
Static	Gray - white matter connectivity	√	0.79 (0.77–0.82)	81.02	80.89	81.16	0.81
Dynamic	Gray matter connectivity	√	0.76 (0.74–0.79)	78.13	77.59	78.70	0.79
Dynamic	White matter connectivity	√	0.75 (0.73–0.78)	77.19	77.76	76.58	0.78
Dynamic	Gray - white matter connectivity		0.82 (0.79–0.84)	83.91	84.68	83.10	0.84
Dynamic	Gray - white matter connectivity	√	0.85 (0.83–0.87)	87.49	87.31	87.68	0.88

Ablation study

To systematically evaluate the efficacy of the input connectivity representations and the contribution of the proposed components, we conducted a series of comprehensive ablation experiments. All experiments were performed under identical conditions, including the same Mamba-CNN backbone, unified data partitioning, consistent hyperparameter settings, and identical evaluation metrics. First, to investigate the advantage of modeling temporal dynamics, we compared the performance of static functional connectivity and dynamic functional connectivity using the same Mamba-CNN architecture. Static functional connectivity was computed as the Pearson correlation matrix across the entire rs-fMRI time series (a single 82×48 matrix per subject), whereas dynamic functional connectivity was derived via sliding-window analysis, yielding a sequence of matrices per subject. Second, to disentangle the respective contributions of different tissue-specific connectivity modalities, we compared three input configurations using the full model architecture: (1) Gray matter connectivity: functional connectivity within and between 82 Gy matter ROIs (82×82 matrix); (2) White matter connectivity: functional connectivity within and between 48 white matter ROIs (48×48 matrix); (3) Gray matter and white matter connectivity: functional connectivity between gray matter ROIs and white matter ROIs (82×48 matrix). Moreover, we validated the effectiveness of the proposed HMSF module by comparing the full model against a variant with the module removed. Table 4 summarizes the ablation results across these different connectivity representations and model components.

Discussions

In the field of ASD research, the development of brain imaging-based intelligent diagnostic models has long centered on gray matter features as the core exploration direction. Early studies mostly focused on the exploration and application of functional connectivity patterns of gray matter. For instance, Yang [40], Kunda [41], Liu [42], Zeraati [43], Hu [44], and Liu [45] analyzed functional connectivity patterns and developed artificial intelligence models with distinct architectures to investigate ASD diagnosis. In the exploration of applying brain structural features, the study by Dong et al. [47] holds significance for expansion. The team not only verified the

effectiveness of using functional connectivity matrix features alone in ASD diagnosis (AUC = 0.74), but also confirmed through comparative experiments that when fMRI functional features were jointly input into the model along with gray matter and white matter structural features, the diagnostic performance was improved (AUC = 0.77). This result provides key evidence for the feasibility of multimodal brain imaging feature fusion and also lays a foundation for subsequent studies to overcome the limitations of relying solely on gray matter features. Recent neuroimaging studies have expanded from traditional gray matter analysis to the systematic exploration of combined gray matter and white matter networks in the brain. Studies represented by the Wang team have found that ASD children exhibit reduced fractional anisotropy, increased mean diffusivity, and increased radial diffusivity in multiple white matter fiber tracts, such as the left anterior thalamic radiation, cingulum bundle, and superior longitudinal fasciculus. Functional connectivity analysis has shown that the gray matter regions connected to these white matter fiber tracts present abnormal functional connectivity patterns [14]. Yang et al. [15] employed the Support Vector Machine (SVM) algorithm to extract functional connectivity features between gray matter and white matter, and conducted a preliminary exploration to compare these features between high-functioning ASD patients and HCs. The findings revealed that abnormal gray matter and white matter connectivity is closely associated with social communication deficits and restricted, repetitive behaviors. The interplay between gray matter and white matter plays a pivotal role in neurodevelopment, cognitive functions, and psychiatric disorders. Recent breakthroughs in AI, particularly the application of deep learning models in neuroimaging, have provided transformative analytical tools for investigating gray matter and white matter interactions. In this work, we propose an innovative Mamba-CNN-based deep learning framework that captures dynamic spatiotemporal patterns of gray matter and white matter functional connectivity to achieve high-accuracy automated diagnosis of ASD.

Our findings demonstrate that when utilizing dynamic spatio-temporal patterns of gray matter and white matter functional connectivity for the diagnosis of ASD, the deep learning model based on the Mamba-CNN framework exhibits robust and generalizable predictive performance at both the single-site level and in multi-site meta-analysis (Table 2). At the individual site level, the proposed model consistently achieved diagnostic performance metrics exceeding the following thresholds across all participating sites, with results surpassing 84.14% accuracy, 82.35% sensitivity, 81.82% specificity, complemented by an AUC exceeding 0.81 and F1 score above 0.81. Such site-level consistency suggests that the model is not overly sensitive to potential variations in data acquisition protocols or demographic characteristics of participants across sites, which is a common challenge in multi-site neuroimaging studies of ASD. The ablation experiments further elucidate the contribution of our proposed connectivity representation. First, the comparison between static and dynamic functional connectivity reveals that modeling temporal fluctuations in gray matter and white matter connectivity captures critical ASD-sensitive information. This finding supports the use of sliding-window approaches for capturing transient gray matter and white matter interaction patterns. Second, gray matter and white matter connectivity yielded the highest accuracy (87.49%), significantly surpassing both only gray matter (78.13%) and only white matter (77.19%) models. This indicates that gray matter and white matter interactions provide complementary information essential for fully characterizing ASD-related brain network aberrations.

Unlike VGG, ResNet and DenseNet, which rely on localized convolutional kernels to extract spatial features, Mamba-CNN's SSM mechanism excels at modeling long-range dependencies and sequential patterns in connectivity data. This is particularly critical for ASD, where disruptions in long-distance gray matter and white matter interactions are well-documented [8, 10, 11]. By efficiently capturing these global dependencies, Mamba achieves a more nuanced representation of ASD-related functional alterations. In the context of gray matter and white matter dynamic functional network analysis, the Mamba-CNN algorithm demonstrates three distinctive advantages. Firstly, its SSM capability effectively integrates structural priors from sMRI with dynamic functional information from fMRI, overcoming the technical limitations of traditional cross-modal fusion approaches. Secondly, the selective attention mechanism adaptively focuses on critical gray matter and white matter connectivity pathways, while filtering out irrelevant features, thereby significantly improving the specificity of feature extraction. Thirdly, the end-to-end training framework enables rapid processing of large-scale neuroimaging datasets, making high-throughput analysis feasible for clinical applications [10].

In this study, the HMSF module was designed within the Mamba-CNN framework. This hierarchical progressive fusion design enables the Mamba-CNN framework to simultaneously process gray matter and white matter functional connectivity features across different scales, enhancing the model's ability to model dynamic sequences. The high-frequency features at the bottom layer preserve the fine-grained dynamic information of local connections, avoiding the loss of details caused by smoothing with a single long time window. In contrast, the features at the middle and top layers integrate the global relevance of macroscale networks, making up for the shortcomings of local isolation and lack of a global perspective in short-window analysis. Through this design, the model achieves a complete interpretation of abnormal interactions in gray matter and white matter dynamic functional connectivity, spanning from local potential associations to global topological coordination. As shown in Table 4, compared with the original Mamba-CNN model, the introduction of the HMSF module increases the model's AUC in identifying ASD-specific features from 0.83 to 0.88. Additionally, the recognition rate for cases with combined gray matter and white matter abnormalities is improved by 2.63%, enhancing the diagnostic sensitivity of the model. Moreover, the applied BCE loss function not only evaluates classification performance on individual frames but also emphasizes the evolutionary patterns of functional connectivity across temporal sequences. This enhancement enables the model to discriminate between static connectivity patterns and transient dynamic abnormalities. Consequently, BCE significantly improves the model's ability to identify dynamic gray matter and white matter network abnormalities associated with ASD, providing a more exploratory biomarker candidates for clinical diagnosis.

The long-range dependency modeling capability of the Mamba-CNN algorithm, combined with Integrated Gradients [48], was used for an exploratory analysis to identify potential connectivity patterns associated with ASD within the gray matter and white matter networks. The specific steps are as follows: (1) for each ASD subject in the test set, the zero matrix was used as the baseline input, and Integrated Gradients computed importance scores for each connection by integrating gradients along the path from the baseline to the actual input; (2) the resulting connection-wise attribution scores were averaged across all ASD subjects to generate a group-level importance ranking; (3) the importance scores were sorted in descending order to identify the connection patterns with the greatest contribution to ASD classification. Meanwhile, the statistical significance was verified via two-sample *t*-tests, and we employed the Benjamini-Hochberg false discovery rate procedure to control for multiple comparisons, with the false discovery rate set at $q = 0.05$ [49]. Notably, the functional connectivity between the superior longitudinal fasciculus and the dorsal anterior cingulate cortex exhibits significant enhancement, suggesting a potential association between altered neuromodulatory balance in this pathway and the observed deficits in higher-order cognitive regulation in ASD, such as emotional evaluation and behavioral inhibition, in individuals with ASD. Conversely, a marked reduction in connectivity between the corpus callosum and the somatosensory association cortex which likely reflects reduced interhemispheric integration efficiency of sensory information and is potentially correlated with impairments in coordinated multimodal sensory processing. Furthermore, a pronounced increase in connectivity is observed between the visual association cortex and the ventral anterior cingulate cortex, a change that may be closely associated with aberrant visual-emotional associative processing in ASD, such as heightened sensitivity to specific visual stimuli or exaggerated emotional responses. Collectively, these specific alterations in connectivity patterns form a characteristic neuromap of dysregulated neural circuits in ASD, providing crucial insights into the neural substrates of core clinical symptoms, including social communication deficits, sensory hypersensitivity, and executive dysfunction.

This study had several limitations. Firstly, although enhanced functional connectivity between the superior longitudinal fasciculus and the dorsal anterior cingulate cortex, as well as weakened connections between the corpus callosum and the somatosensory association cortex, have been identified, whether these neural connectivity abnormalities directly contribute to the core symptoms of ASD, such as impairments in social communication and restricted, repetitive behaviors, remains to be rigorously validated through well-designed experimental studies. Secondly, the current study is the treatment of ASD as a single homogeneous diagnostic category. The present analysis, by pooling all ASD participants regardless of potential subtypes, may obscure subtype-specific gray matter and white matter connectivity patterns and underestimate the model's potential diagnostic performance.

Recent data-driven subtyping studies have identified multiple ASD subtypes with distinct neuroanatomical and functional patterns [50]. Future work will incorporate multimodal subtyping approaches and stratified analyses across different developmental stages to investigate whether the identified abnormal connectivity patterns are differentially represented across ASD subtypes and age groups. Furthermore, ASD patients exhibit significant clinical heterogeneity, including variations in language proficiency, disparities in intellectual functioning, and a diverse range of comorbid psychiatric disorders. To account for these confounding factors, future research should employ more refined stratification analyses to elucidate the specific associations between neuroimaging features and clinical phenotypes.

Conclusion

In conclusion, this study focused on the dynamic network connectivity between gray matter and white matter in ASD, aiming to address the challenges in analyzing spatio-temporally complex neuroimaging data and explore novel technical pathways for elucidating the neural mechanisms of ASD. This study proposed an innovative deep learning framework based on the Mamba architecture, effectively addressing technical bottlenecks in dynamic brain network analysis. The deep learning method achieved a diagnostic accuracy of 87.49% for ASD, outperforming other artificial intelligence approaches. By integrating neuroimaging data analysis and ASD neurobiological research, we identified unique patterns of abnormal dynamic interactions in the brain networks of ASD patients. These findings not only complement the existing understanding of gray matter and white matter connectivity abnormalities in ASD but also provide a new integration perspective for deciphering the neural mechanisms underlying ASD's core symptoms.

Abbreviations

ASD	Autism spectrum disorder
sMRI	Structural magnetic resonance imaging
fMRI	Functional magnetic resonance imaging
HCs	Healthy controls
AI	Artificial intelligence
SSMs	State space models
TR	Repetition time
FD	Framewise displacement
ROI	Region of interest
VSS	Visual state space
SS2D	2D selective scan
HMSF	Hierarchical multi-scale fusion
MLP	Multi-layer perceptron
BCE	Binary cross entropy
ACC	Accuracy
SEN	Sensitivity
SPE	Specificity
AUC	Area under the curve
ROC	Receiver operating characteristic
SVM	Support vector machine

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Author contributions X.Y. designed the study, developed the protocol, collected data, reviewed literature, and contributed to writing the manuscript. Y.W., Y.B., S.J. contributed to design and construct the model. Y.S., J.T. oversaw data analysis. NM reviewed literature, collected data and revised the important content critically. Y.K., X.Z. contributed technical support. Y.L., H.X. oversaw data analysis and processing. M.W. oversight the research activity execution and acquired the financial support for the project. All authors read and approved the final manuscript.

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Data availability The ABIDE dataset is a publicly available, fully anonymized database. This study was performed in line with the principles of the Declaration of Helsinki. All research procedures involving human participants conformed to the ethical standards of the Research Committee of the Henan Provincial People's Hospital (approval number: IRB 2023–104).

Declarations

Ethics and consent to participate declarations This study was performed in line with the principles of the Declaration of Helsinki. All research procedures involving human participants conformed to the ethical standards of the Research Committee of the Henan Provincial People's Hospital (approval number: IRB 2023–104). Written informed consent was obtained from a parent/guardian, and assent was obtained from each child.

Consent for publication Not applicable.

Competing interests The authors declare no competing interests.

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